

Report on Model Evaluation & Refinement
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Project: Gachibowli Traffic Prediction

1. MODEL DEVELOPMENT AND TRAINING:

- Dataset: 2184 rows (Jan-Mar 2025 hourly), Peak 9AM 722, 6PM 738
- Time-based Split: 80% Train (Jan1-Mar13 Past) / 20% Val (Mar13-Mar31 Future)
- ARIMA(5,1,0) for time series forecasting
- LSTM: 24hr sequence (1723,24,1) for sequence handling
- Gradient Boosting: Robust & Interpretable
- Grid Search Best: lr=0.05, depth=3, n_est=200, MAE 27.12
- Random Search: randint, uniform, 5 iter

2. MODEL EVALUATION & CROSS VALIDATION:

- Metrics: MAE 26.15, RMSE, R2 0.97 (97% variance)
- Validation on Future Data (Mar13-31)
- Visuals: Prediction vs Actual, Residual Plot, Error Distribution
- k-Fold CV (5 folds): MAE 27.14 $\pm$ 0.63 Stable
- Time-Based CV (3 folds, temporal order): 27.30 $\pm$ 0.23 Very Consistent
- Analysis: Low std = Consistent, Train 25.70 vs Val 26.15 diff 0.45 = GOOD GENERALIZATION

3. MODEL REFINEMENT:

- Diagnosis: Low bias/variance, Systematic error at peak hours 9AM/6PM
- Error Analysis: Rain MAE > Clear, Peak hours higher error
- New Features: Lag_1h, Lag_24h, Rolling_3h_Mean, IsPeakAM/PM
- Removed: Junction (redundant constant)
- Algorithms: GB 24.31, RF 25.24, Stacking Ensemble (GB+RF+Ridge) 24.24 BEST
- Grid Search Refined: lr=0.1, max_depth=4, CV MAE 25.66, Final 24.37, R2 97.22%

4. FINAL MODEL:

- Stacking Ensemble, MAE 24.24 -> 24.37, Improvement 2.9 vehicles
- Files: .ipynb, Final Predictions CSV, This PDF
- Conclusion: Time-based + ensemble + lag features = Ready for deployment